

Medical Image Segmentation and Applications

Lab- P1: The SPM Framework

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1. Introduction and problem definition

The objective of this experiment is to get introduced to a commonly used neuroimaging framework known as Statistical Parametric Mapping (SPM) and to apply brain tissue segmentation for a given dataset. A dataset containing brain MRI images of five different patients are given. Our aim is to apply the two most common brain pre-processing steps: a) Bias Field Correction and b) Skull stripping process. We used the SPM toolkit in MATLAB software to obtain the brain tissue segmentation (WM, GM, CSF) and then evaluated our outcomes with the given ground truth labels based on dice score. The working procedure of the experiment and results are discussed in the next sections.

2. Design and Implementation

To conduct the experiment, we first installed the SPM toolkit in our Matlab software. Using the command “spm” in matlab terminal we run the SPM toolkit. For brain tissue segmentation, we chose fMRI ► Segment option. Before performing the tissue segmentation, we set the different parameters like bias regularization, bias FWHM, number of gaussians, warped tissues, MRF parameters. Setting different parameter values gives us different segmentation results.

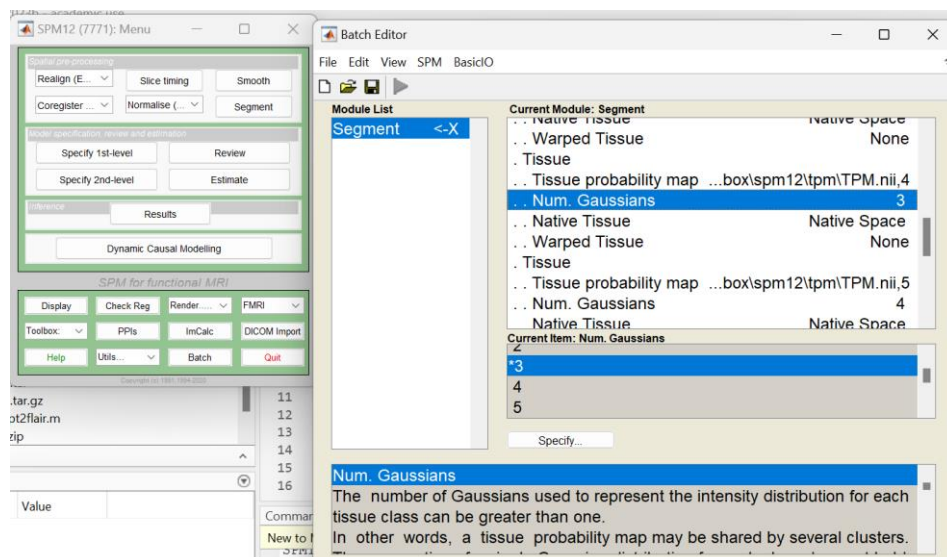


Fig-1: SPM Toolkit for Segmentation

Then for each patient data, we performed Bias field removal and generated the 6 probe maps (Grey matter, White matter, CSF, bone, white tissue, AIR). The generated

probe maps were in “.nii” format hence we visualized them using the ITK-Snap toolkit.

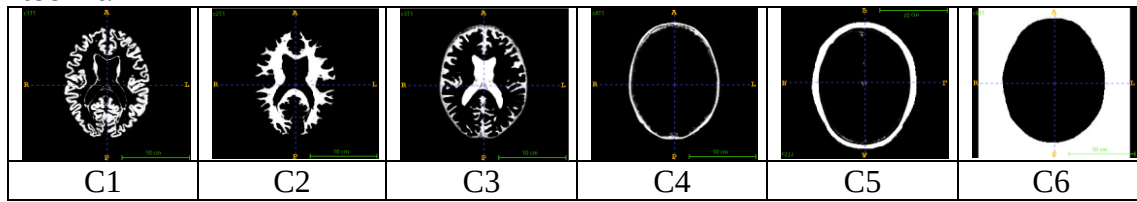


Fig-2: Generated probe maps form patient 1 MRI image

After performing the first preprocessing step of bias field removal and generating the probe maps we performed skull stripping. To implement skull stripping, we first combined GM, WM and CSF to create a brain mask and then we multiplied this brain mask with the original image data. We performed the skull stripping by writing the code on MATLAB. A snippet of the code is shown below.

```
gm = spm_read_vols(spm_vol(gm_file));
wm = spm_read_vols(spm_vol(wm_file));
csf = spm_read_vols(spm_vol(csf_file));

% Combine GM, WM, and CSF to create the brain mask
brain_mask = gm + wm + csf;
brain_mask = brain_mask > 0; % Binary mask (0 for non-brain, 1 for brain)

% Apply the mask to the original image to get the skull-stripped image
skull_stripped_img = img .* brain_mask;
```

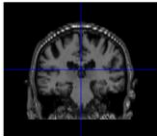
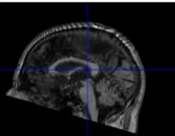
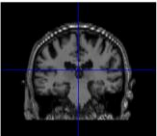
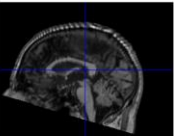
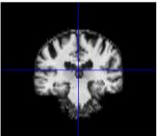
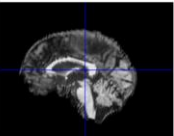
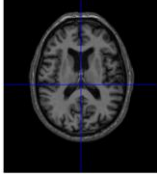
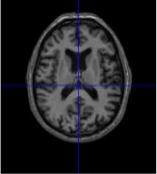
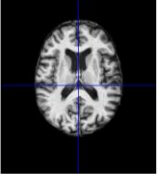
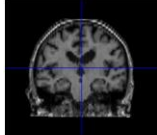
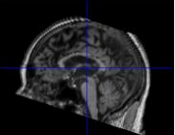
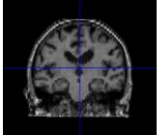
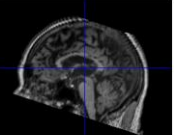
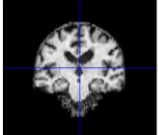
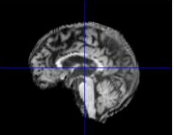
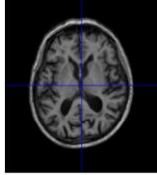
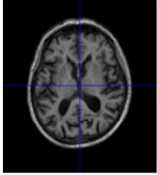
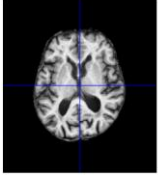
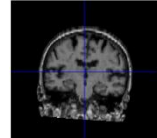
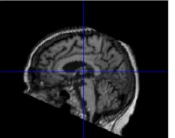
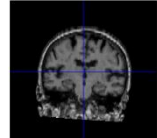
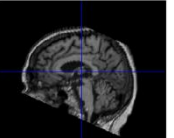
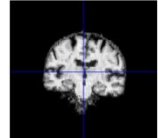
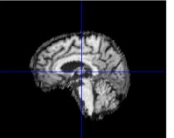
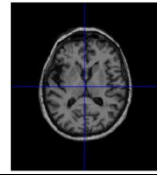
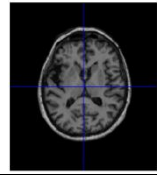
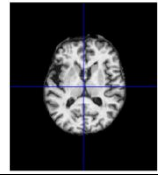
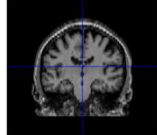
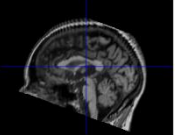
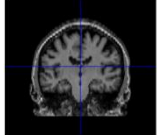
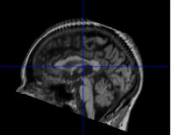
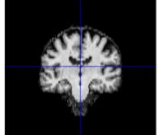
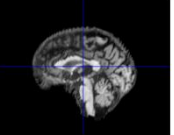
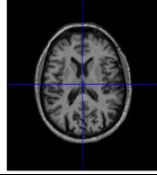
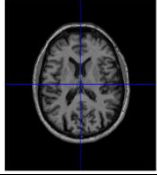
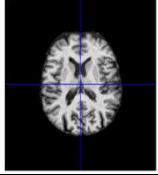
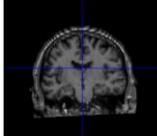
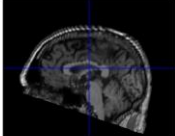
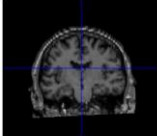
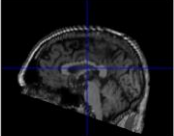
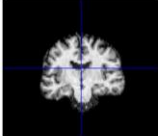
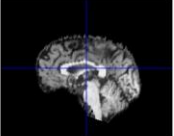
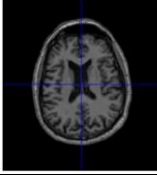
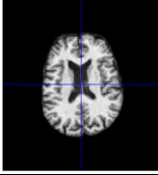
The labelled ground truth of each patient data was also provided. Hence, to evaluate the segmentation process, we calculated the dice score coefficients of GM, WM and CSF comparing with the ground truth labels. This was also implemented using MATLAB code. Finally for further evaluation we calculated the mean and standard deviation of the dice scores.

3. Experimental Section and Result Analysis

The given dataset consists of MRI brain images of five patients. For each patient data the ground truth is also provided. The preprocessing results (Bias Field Removal and Skull Stripping Process) for each patient data are illustrated in figure-3.

Patient Number	Data Files
1	T1.nii T2_FLAIR.nii LabelsForTesting.nii
2	T1.nii T2_FLAIR.nii LabelsForTesting.nii
3	T1.nii T2_FLAIR.nii LabelsForTesting.nii
4	T1.nii T2_FLAIR.nii LabelsForTesting.nii
5	T1.nii T2_FLAIR.nii LabelsForTesting.nii

Table-1: Dataset Description

Original Image		Bias Field Corrected		Skull Stripped	
					
					
Patient-1: T1.nii					
					
					
Patient-2: T1.nii					
					
					
Patient-3: T1.nii					
					
					
Patient-4: T1.nii					
					
					
Patient-5: T1.nii					

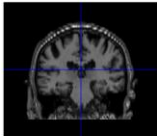
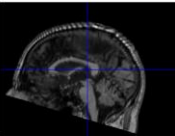
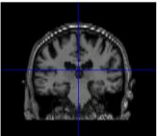
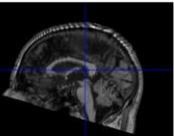
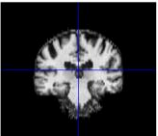
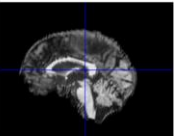
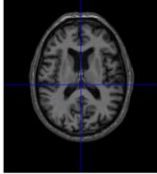
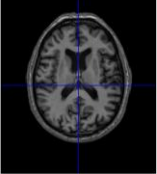
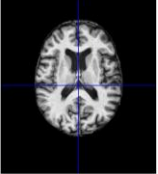
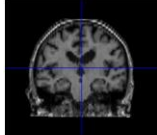
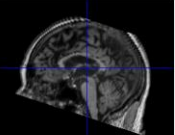
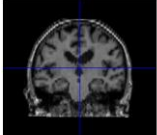
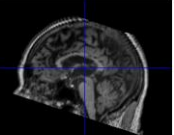
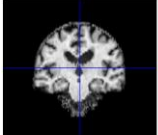
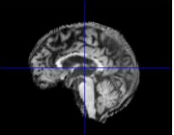
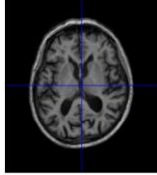
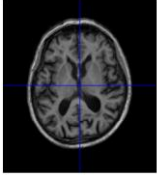
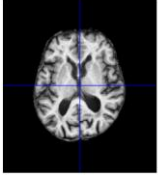
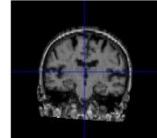
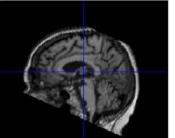
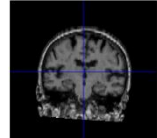
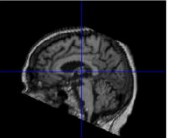
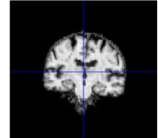
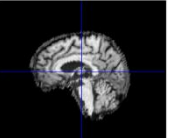
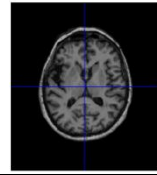
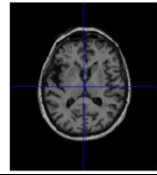
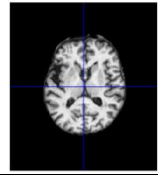
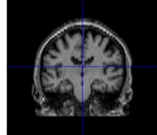
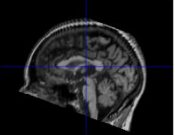
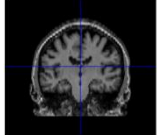
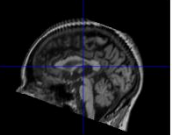
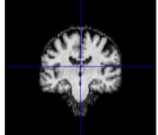
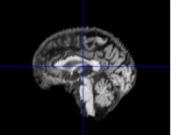
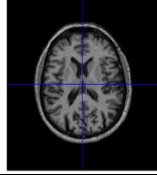
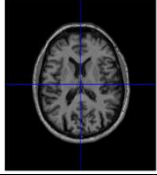
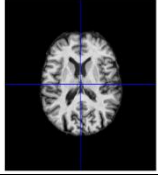
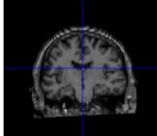
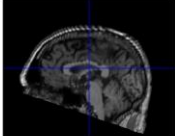
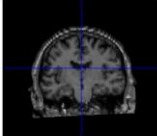
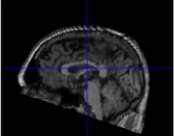
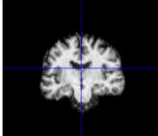
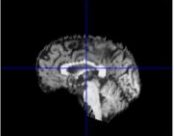
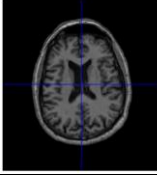
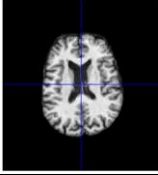
Original Image		Bias Field Corrected		Skull Stripped	
					
					
Patient-1: T2_Flair.nii					
					
					
Patient-2: T2_Flair.nii					
					
					
Patient-3: T2_Flair.nii					
					
					
Patient-4: T2_Flair.nii					
					
					
Patient-5: T2_Flair.nii					

Figure-3: Original and Processed Brain Tissue segmentation of each patient data

The segmented images are compared with the given ground label truth and evaluated based on dice scores for WM, GM and CSF. The results are shown in table-2 and table-3 respectively.

Patient Number	Test Data	Segmented Tissues	Dice Scores
Patient 1	T1	GM	0.6691
		WM	0.7979
		CSF	0.6339
	T2_Flair	GM	0.6114
		WM	0.5176
		CSF	0.6055
Patient 2	T1	GM	0.7001
		WM	0.6962
		CSF	0.4913
	T2_Flair	GM	0.6279
		WM	0.5796
		CSF	0.4428
Patient 3	T1	GM	0.6467
		WM	0.8065
		CSF	0.5694
	T2_Flair	GM	0.5629
		WM	0.4971
		CSF	0.4319
Patient 4	T1	GM	0.7104
		WM	0.7664
		CSF	0.6872
	T2_Flair	GM	0.6472
		WM	0.4601
		CSF	0.5803
Patient 5	T1	GM	0.7086
		WM	0.8045
		CSF	0.6271
	T2_Flair	GM	0.6755
		WM	0.4817
		CSF	0.4756

Table-2: Dice Scores of segmented tissues

Patient Number	Test Subjects	Segmented Tissues	Mean	Standard Deviation
Patient 1-Patient 5	T1	GM	0.6870	0.0280
		WM	0.7743	0.0466
		CSF	0.6018	0.0745
	T2_Flair	GM	0.6250	0.0421
		WM	0.5072	0.0456
		CSF	0.5072	0.0803

Table-3: Mean and Standard Deviation of Dice scores

The results showed that the segmented White Matter probe gives comparatively better results than the other two segmented tissues. The results can be further improved by tuning the different parameters in the SPM toolkit while segmenting.

4. Project Management details

In this project, the primary tasks involved installing and configuring the SPM toolkit, performing brain tissue segmentation, applying bias field correction, and skull stripping. Time estimation for setting up the SPM environment and understanding its functionality was approximately 2 hours, while implementing tissue segmentation and preprocessing for each patient's MRI data was estimated to take another 3 hours. The actual time dedication for these tasks, however, extended to about 7 hours due to troubleshooting issues during installation and parameter tuning in segmentation. Evaluating the Dice scores and comparing results with ground truth data was initially planned for 1 hour but required 2 hours due to unexpected discrepancies in segmentation accuracy.

5. Conclusion

In this experiment, the SPM framework was successfully applied to perform brain tissue segmentation, including preprocessing steps such as bias field correction and skull stripping. The segmentation process generated Grey Matter (GM), White Matter (WM), and Cerebrospinal Fluid (CSF) tissue maps, which were compared against ground truth labels using Dice similarity scores. The results revealed that the segmentation of White Matter consistently produced higher Dice scores, particularly with T1 images, showing a mean Dice score of 0.7743. However, Grey Matter and CSF segmentation results were less accurate, with lower Dice scores across the dataset.

Through this lab experiment, we learned how to use the SPM toolkit and perform basic pre-processing tasks which will be helpful to perform high level processing in the subsequent labs.